

**Innovative Mathematics Olympiad
Model Paper 2025 – 2026**

Total Questions: 35

Time: 1 hr.

Total Marks: 70

Syllabus

Number Sense – Number Names (5-digit & 6-digit numbers), Comparing and Formation of Numbers, Basic Operations (+, −, ×, ÷), Factors and Multiples, Fractions and Decimals, Time and Money, Measurement of Length, Area, Mass, Volume, Geometry – Plane Geometry & Symmetry, Data Handling.

Instructions:

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

MATHEMATICS

1. A number is formed using digits 2, 4, 5, 7 without repetition. What is the greatest 4-digit number divisible by 3?
(A) 9752 (B) 7542 (C) 7592 (D) 7952
2. The estimated product of $4,892 \times 61$ using rounding is:
(A) 2,90,000 (B) 3,00,000 (C) 3,10,000 (D) 2,80,000
3. Statement I: Every multiple of 9 is divisible by 3.
Statement II: If the sum of digits of a number is divisible by 3, then the number is divisible by 3.
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true
4. What is the HCF of the smallest composite number and the largest one-digit prime number?
(A) 1 (B) 2 (C) 3 (D) 4
5. Which fraction is equivalent to $\frac{18}{24}$ and in its simplest form?
(A) $\frac{6}{8}$ (B) $\frac{3}{4}$ (C) $\frac{9}{12}$ (D) $\frac{12}{16}$

6. A train departs at 18:40 (24-hour format) and reaches its destination after 2 hours 35 minutes. Arrival time is:
 (A) 21:05 (B) 20:75 (C) 21:15 (D) 20:15
7. The perimeter of a square is equal to the perimeter of a rectangle of length 14 cm and breadth 10 cm. Side of the square is:
 (A) 10 cm (B) 12 cm (C) 14 cm (D) 24 cm
8. Which of the following figures must always be a closed figure?
 (A) Ray (B) Line (C) Polygon (D) Angle
9. A pictograph shows  = 4 cars. If 9 symbols are shown, total cars are:
 (A) 13 (B) 32 (C) 36 (D) 40
10. Which number leaves remainder 0 when divided by both 6 and 8?
 (A) 12 (B) 24 (C) 36 (D) 46
11. $\frac{3}{5}$ of a number is 48. The number is:
 (A) 72 (B) 80 (C) 60 (D) 96
12. A number when rounded to the nearest thousand becomes 54,000. Which of the following cannot be the number?
 (A) 53,501 (B) 54,499 (C) 54,501 (D) 53,750
13. The value of $6^2 + 4 \times 3$ is:
 (A) 48 (B) 36 (C) 30 (D) 24
14. Which of the following numbers has exactly three factors?
 (A) 8 (B) 9 (C) 10 (D) 15
15. Which of the following is the largest?
 (A) $\frac{3}{4}$ (B) $\frac{5}{8}$ (C) $\frac{7}{10}$ (D) $\frac{9}{16}$
16. A movie starts at 3:45 PM and ends at 6:10 PM. What is the duration of the movie?
 (A) 2 h 15 min (B) 2 h 25 min (C) 2 h 35 min (D) 2 h 45 min
17. The area of a rectangle is 432 cm² and its length is 27 cm. The breadth is:
 (A) 12 cm (B) 14 cm (C) 16 cm (D) 18 cm
18. Which statement is always true?
 (A) All quadrilaterals are squares
 (B) All rectangles are parallelograms
 (C) All parallelograms are rectangles
 (D) All triangles are isosceles

19. Which of the following is correctly matched?

Quantity		Standard Unit
(A) Length	—	Kilogram
(B) Mass	—	Meter
(C) Volume	—	Liter
(D) Area	—	Meter

20. The scale of a pictograph is  = 5 bicycles. If 7 symbols are shown, total bicycles are:

- (A) 30 (B) 32 (C) 35 (D) 4

21. Which is the smallest number divisible by both 9 and 12?

- (A) 18 (B) 24 (C) 36 (D) 72

22. What is the decimal form of $\frac{7}{8}$?

- (A) 0.875 (B) 0.78 (C) 0.825 (D) 0.887

23. If 15 notebooks cost ₹180, what is the cost of 8 notebooks?

- (A) ₹88 (B) ₹90 (C) ₹96 (D) ₹100

24. A number has digits 4, 5, 6. How many 3-digit even numbers can be formed without repetition?

- (A) 1 (B) 2 (C) 3 (D) 4

25. Which of the following is not divisible by 6?

- (A) 324 (B) 510 (C) 418 (D) 672

26. The sum of two unlike fractions is $\frac{11}{12}$. One fraction is $\frac{1}{3}$. The other fraction is:

- (A) $\frac{5}{12}$ (B) $\frac{7}{12}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$

27. A shopkeeper buys a table for ₹2,400 and sells it at a loss of ₹180. The selling price is:

- (A) ₹2,220 (B) ₹2,200 (C) ₹2,180 (D) ₹2,260

28. The perimeter of an equilateral triangle is 45 cm. Each side is:

- (A) 12 cm (B) 13 cm (C) 15 cm (D) 18 cm

29. Which shape has exactly one line of symmetry?

- (A) Square (B) Rectangle
(C) Isosceles triangle (D) Circle

30. A bar graph uses a scale of 1 unit = 10 students. A bar of height 7 units represents:

- (A) 17 students (B) 70 students
(C) 700 students (D) 107 students

31. Which of the following is a prime factor of 132?

- (A) 4 (B) 6 (C) 11 (D) 22

32. What is the value of $0.6 + 0.75$

- (A) 1.25 (B) 1.35 (C) 1.45 (D) 1.55

33. A bus covers 180 km in 3 hours. At the same speed, distance covered in 5 hours is:

- (A) 250 km (B) 280 km (C) 300 km (D) 320 km

34. Which of the following numbers has exactly two factors?

- (A) 21 (B) 29 (C) 39 (D) 49

35. The simplest form of $\frac{84}{126}$ is:

- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) $\frac{6}{9}$ (D) $\frac{14}{21}$

KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) B | 2) B | 3) A | 4) A | 5) B | 6) C | 7) B |
| 8) C | 9) C | 10) B | 11) B | 12) C | 13) A | 14) B |
| 15) A | 16) B | 17) C | 18) B | 19) C | 20) C | 21) C |
| 22) A | 23) C | 24) D | 25) C | 26) B | 27) A | 28) C |
| 29) C | 30) B | 31) C | 32) B | 33) C | 34) B | 35) A |